

Multi-Directional Smart Bicycle Blind Spot & Collision Alert System



Presented To:

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Problem Statement

- Cyclists cannot see blind spots easily so high risk of collision from back and sides
- No affordable smart safety system for people



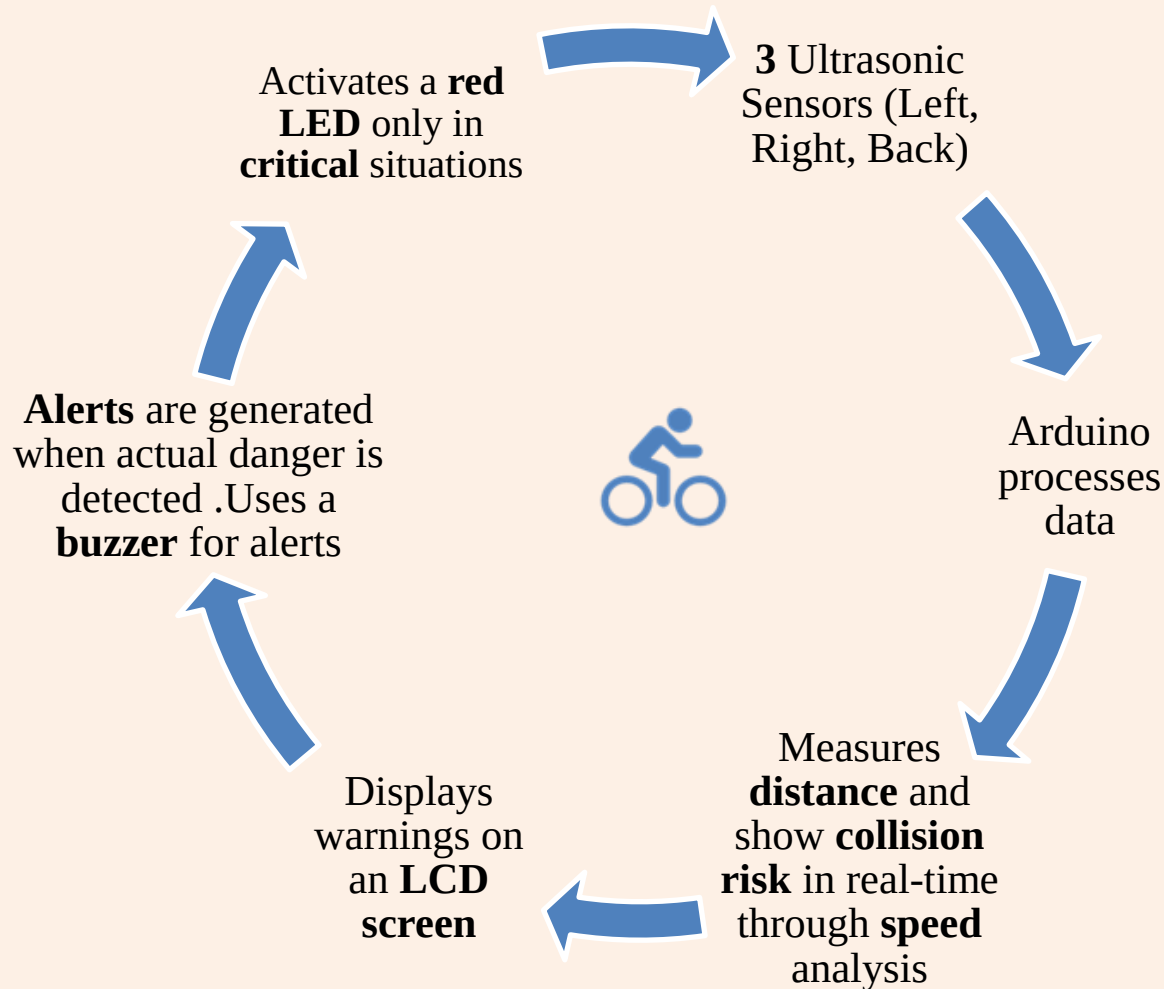
Project Idea

“A multi-directional collision alert system designed for bicycle and rickshaw riders that **detects** vehicles approaching from the left, right, and rear sides and provides **real-time** warnings using LCD, buzzer, and a **critical** LED alert ”

Who will Benefit:

- Student cyclists
- Delivery riders such as FoodPanda, Pathao riders
- Rickshaw pullers

Project Overview



Key Features

1. Multi-Directional Detection System:

- i. 3 ultrasonic sensors (Left, Right, Back)
- ii. Covers all blind spots and provides full directional awareness

2. Real-Time Distance Analysis :

- i. Measures distance in centimeters
- ii. Detects closest vehicle and prioritizes highest risk direction

3. Speed Approximation:

- i. Calculates speed using distance change
- ii. Fast decrease → High risk and Slow decrease → Normal condition

4. Smart LCD Display (MAIN OUTPUT) :

- i. Displays all system information
- ii. Shows direction and distance clearly and provides real-time safety messages

Key Features

5. Buzzer Alert System:

Condition	Buzzer
Safe	OFF
Warning	Slow beep
High risk	Fast beep
Critical	Continuous loud sound

6. Critical Safety LED:

- i. Activates only in critical condition which indicates emergency situation
- ii. Red LED ON + continuous buzzer

7. False Alert Reduction System

- i. Uses multiple sensor readings and averages data for accuracy
- iii. Ignores sudden spikes

Output States (LCD Display)

1. SAFE STATE:

All Clear.

L: Safe | R: Safe | B: Safe

2. WARNING :

Vehicle approaching

Shows direction + distance

3. HIGH RISK :

Danger Zone ! Vehicle very close !

Shows direction + distance

4. CRITICAL :

STOP IMMEDIATELY !!!

Collision Risk Detected

Shows direction + distance

Components

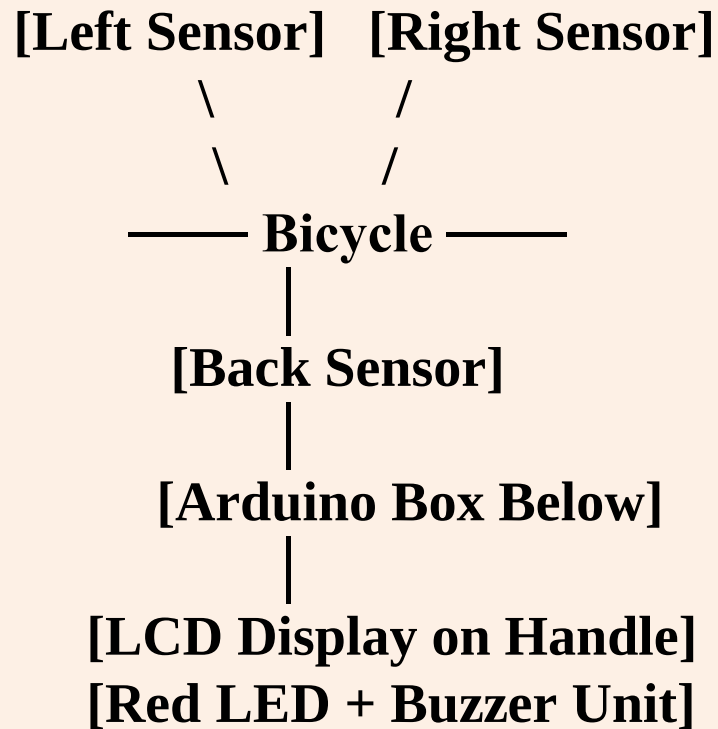
1. Arduino Uno / Nano
2. HC-SR04 Ultrasonic Sensors ×3
3. LCD 16×2 Display
4. Buzzer Red LED
5. Battery / Power Bank
6. Toy Bicycle Model

Estimated Cost

Total Cost: Tk1500 – Tk 2000 (approx.)

Low-cost and affordable solution & Suitable for mass use

Physical Design



Potential Risks / Challenges

- Ultrasonic sensors may be affected by rain, dust, or uneven road surfaces
- False alerts may occur in crowded traffic environments
- Detection accuracy can reduce at high vehicle speeds
- Battery life and power management need optimization



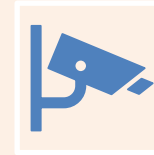
Future Advancement



Integrating AI for smarter object and risk prediction



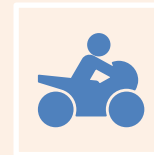
Adding GPS and mobile app connectivity for tracking & alerts



Using camera/radar sensors for higher accuracy



Developing waterproof and compact real-world design



Expanding system for motorcycles and other vehicles

Conclusion

- Multi-directional real-time collision alert system
- Simple, low-cost, and effective for road safety
- Reduces blind spot risks and improves rider awareness
- Suitable for future real-world development

